

Inferential Seamless Designs for Treatment Selection during a Phase II/III Study in Oncology

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Project Information

Duration: July 01th 2024 - December 20th 2024

Master Thesis

Objective: Efficient Phase II/III MAMS design for optimal dose/drug combination.

Focus:

- Drop-the-Losers approach (Wason et al., 2017)
- Strong advantages vs other MAMS Design
- Smaller sample sizes, higher power, reduced duration
→ Cost Reduction, Time-to-Market
- Test the assumption with a simulation study

Goal: Extension of the concept to **binary** and **survival endpoints**

Article

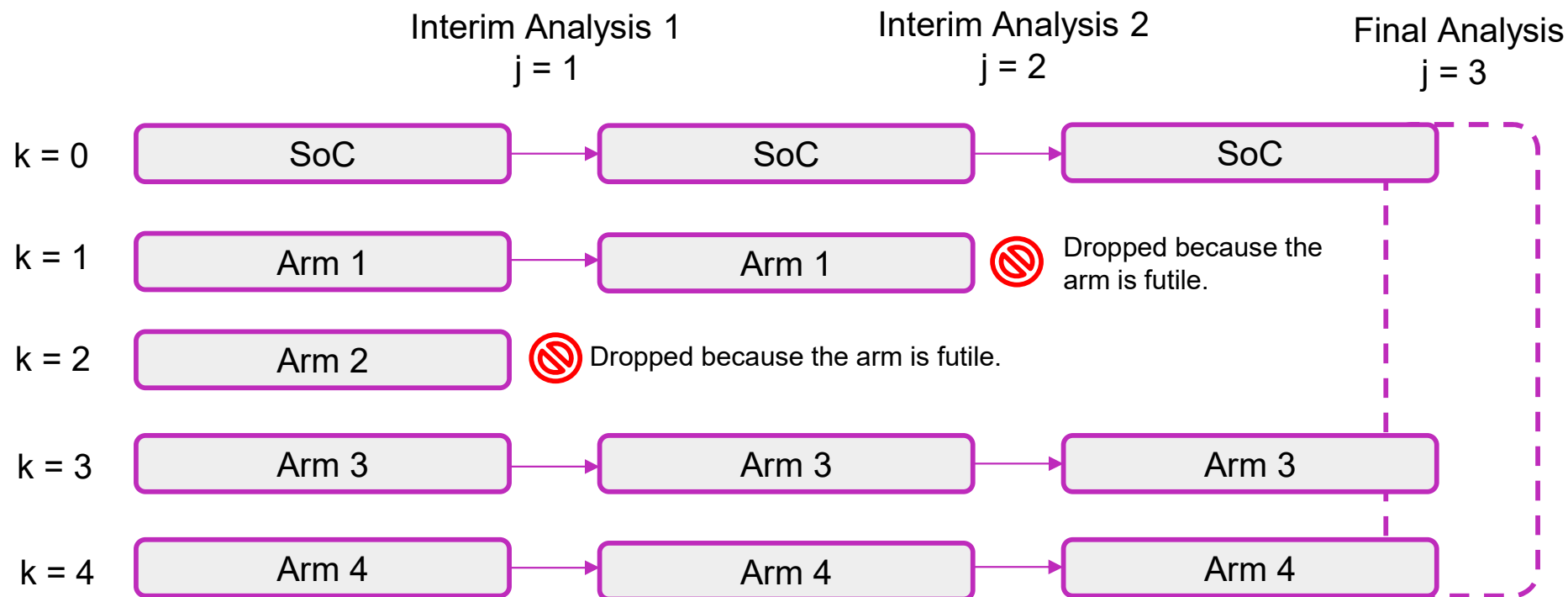
A multi-stage drop-the-losers design for multi-arm clinical trials

James Wason,¹ Nigel Stallard,² Jack Bowden¹
and Christopher Jennison³

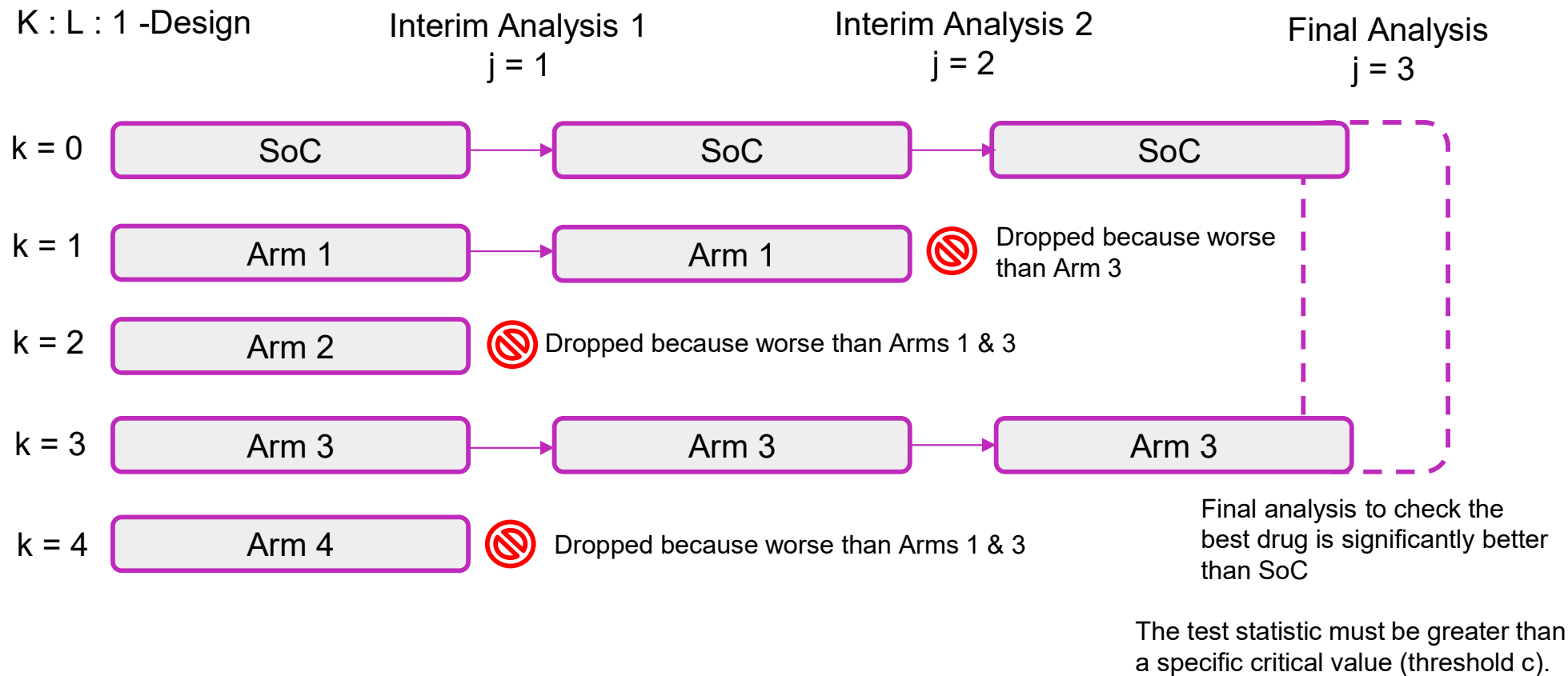


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Definition: MAMS trials, multiple experimental treatments tested against a control in the same trial.



Drop-the-Losers Design



Drop-the-Losers Design

Advantages:

- **Study Cost & Sample Size:** The sample sizes are fixed, allowing for precise cost estimation.
- **Flexibility:** adaptable to interim analyses and continuous monitoring, with the ability to remove ineffective arms, offering advantages over independent trials (parallel)
- **Reduced administrative burden:** only one placebo group used; single-arm trials require individual study approvals.
- **Practical Applications:** Ideal for trials with multiple arms, quick elimination of ineffective treatments, advantage over traditional designs.

Limitations:

- Computational complexity, error control challenges, difficulty with small effect sizes, complex calculations, robust planning needed.

Continuous Endpoints

$$Z_j^{(k)} = \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\sigma^2}}$$

Assume $p > j$ and $n_p > n_j$ and $k = m$

We want to solve:

$$\text{Cov} \left(Z_j^{(k)}; Z_p^{(k)} \right)?$$

Notation:
 j & p for analyses
 k & m for arms
 X for Patient's Outcome

X_{kj} : Outcomes for treatment group k at stage j .
 X_{0j} : Outcomes for control group at stage j .
 n_j : Total number of observations at stage j .
 σ^2 : Known variance.

Source: Wason et al., 2017

$$\begin{aligned} \text{Cov}(Z_j^{(k)}; Z_p^{(k)}) &= \text{Cov} \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \sqrt{\frac{n_j}{2\sigma^2}}; \frac{\sum_{i=1}^{n_p} X_{ki}}{n_p} - \frac{\sum_{i=1}^{n_p} X_{0i}}{n_p} \sqrt{\frac{n_p}{2\sigma^2}} \right) \\ &= \frac{\sqrt{n_j n_p}}{2\sigma^2} \cdot \text{Cov} \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}; \frac{\sum_{i=1}^{n_p} X_{ki}}{n_p} - \frac{\sum_{i=1}^{n_p} X_{0i}}{n_p} \right) \\ &= \frac{\sqrt{n_j n_p}}{2\sigma^2} \left[\text{Cov} \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j}; \frac{\sum_{i=1}^{n_p} X_{ki}}{n_p} \right) - \text{Cov} \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j}; \frac{\sum_{i=1}^{n_p} X_{0i}}{n_p} \right) \right. \\ &\quad \left. - \text{Cov} \left(\frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}; \frac{\sum_{i=1}^{n_p} X_{ki}}{n_p} \right) + \text{Cov} \left(\frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}; \frac{\sum_{i=1}^{n_p} X_{0i}}{n_p} \right) \right] \\ &= \frac{\sqrt{n_j n_p}}{2\sigma^2} \left[\text{Cov} \left(\frac{1}{n_j} \sum_{i=1}^{n_j} X_{ki}; \frac{1}{n_p} \sum_{i=1}^{n_p} X_{ki} \right) \right. \\ &\quad \left. + \text{Cov} \left(\frac{1}{n_j} \sum_{i=1}^{n_j} X_{0i}; \frac{1}{n_p} \sum_{i=1}^{n_p} X_{0i} \right) \right] \\ &= \frac{\sqrt{n_j n_p}}{2\sigma^2} \left[\frac{1}{n_j n_p} \sum_{i=1}^{n_j} \text{Cov}(X_{ki}, X_{ki}) + \frac{1}{n_j n_p} \sum_{i=1}^{n_j} \text{Cov}(X_{0i}, X_{0i}) \right] \\ &= \frac{\sqrt{n_j n_p}}{2\sigma^2} \left[\frac{1}{n_j n_p} \cdot 2n_j \sigma^2 \right] = \sqrt{\frac{n_j}{n_p}} \end{aligned}$$

Continuous Endpoints

$$Z_j^{(k)} = \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\sigma^2}}$$

Assume $p > j$ and $n_p > n_j$ and $k \neq m$:

We want to solve:

$$\text{Cov} \left(Z_j^{(k)}; Z_p^{(k)} \right)?$$

Notation:

j & p for analyses

k & m for arms

X for Patient's Outcome

$$\text{Cov} \left(Z_j^{(k)}; Z_p^{(k)} \right) = \frac{1}{2} \cdot \sqrt{\frac{n_j}{n_p}}$$

Contrast Matrix A

$$\begin{array}{l} \text{IA1} \\ \text{IA2} \\ \text{FA} \end{array} \left\{ \mathbf{A} = \begin{array}{c} \text{Arm 1} \quad \text{Arm 2} \quad \text{Arm 3} \quad \text{Arm 4} \\ \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \end{array} \right. \mathbf{z} = \begin{pmatrix} Z_1^1 \\ Z_2^1 \\ Z_3^1 \\ Z_1^2 \\ Z_2^2 \\ Z_3^2 \\ Z_1^3 \\ Z_2^3 \\ Z_3^3 \\ Z_1^4 \\ Z_2^4 \\ Z_3^4 \end{pmatrix} \quad AZ \sim N(A\mu, A\Sigma A^T)$$

Probability of a specific treatment being recommended:

$$\psi = (\psi_1, \dots, \psi_K)$$

We introduce random variables,

where ψ_k is the ranking of treatment k.

$$P(\text{Reject } H_0^{(k)} | \delta) = P(\psi_k = 1, Z_3^{(k)} > c | \delta)$$

Ranking:

- (1) the treatment that reaches the final analysis has rank 1;
- (2) the treatment that is dropped at the first analysis with the lowest test statistic is given rank, K ;
- (3) if treatment k_1 reaches a later stage than treatment k_2 , then $\psi_{k_1} < \psi_{k_2}$, that is, treatment k_1 has a higher ranking;
- (4) if treatments k_1 and k_2 are dropped at the same stage, and k_1 has a higher test statistic at that stage, then $\psi_{k_1} < \psi_{k_2}$.

Source: Wason et al., 2017

Global Null Hypothesis:

Definition: The global null hypothesis assumes that none of the treatments are more effective than the control.

Formally: $\mu_1 = \mu_2 = \dots \mu_4 = \mu_{control}$

Global Alternative Hypothesis:

Definition: The global alternative hypothesis posits that at least one treatment is significantly better than the control.

Formally: $H_A \exists i \text{ such that } \mu_i > \mu_{control}$

Expected mean effect: $m(\delta)$

The expected mean effect $m(\delta)$ represents the anticipated average treatment effect for each arm compared to the control group.

Probability of recommending any treatment under the global null hypothesis:

Under global null hypothesis H_0 : each element $m(\delta) = 0$. Symmetry: equal probability of ordering ψ and $Z > c$. Probability of treatment recommendation

$$K! \mathbb{P}(\psi_1 = 1, \psi_2 = 2, \dots, \psi_K = K, Z_J^{(1)} > c | \delta = 0)$$

Probability of recommending a specific treatment under the least favorable configuration:

$$(K-1)! \mathbb{P}(\psi_1 = 1, \psi_2 = 2, \dots, \psi_K = K, Z_J^{(1)} > c | \delta_1 = \delta^{(1)}, \delta_2 = \delta^{(0)}, \dots, \delta_K = \delta^{(0)})$$

Source: Wason et al., 2017

$$Z_j^{(3)} = \left(\frac{\sum_{i=1}^{n_j} X_{3i}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\sigma^2}}$$

X_{kj} : Outcomes for treatment group k at stage j .

X_{0j} : Outcomes for control group at stage j .

n_j : Total number of observations at stage j .

σ^2 : Known variance.

$$Z_j^{(3)} \sim N\left(\delta_3 \sqrt{\frac{n_j}{2\sigma^2}}, 1\right)$$

$$\delta_3 = \mu_3 - \mu_0$$

The null hypotheses to be tested are $H_0^3 : \delta_3 \leq 0$.

The global null hypothesis, H_G , is defined as $H_G : \delta_1 = \delta_2 = \dots = \delta_k = 0$

Results from Wason Paper

Drop-the-Losers

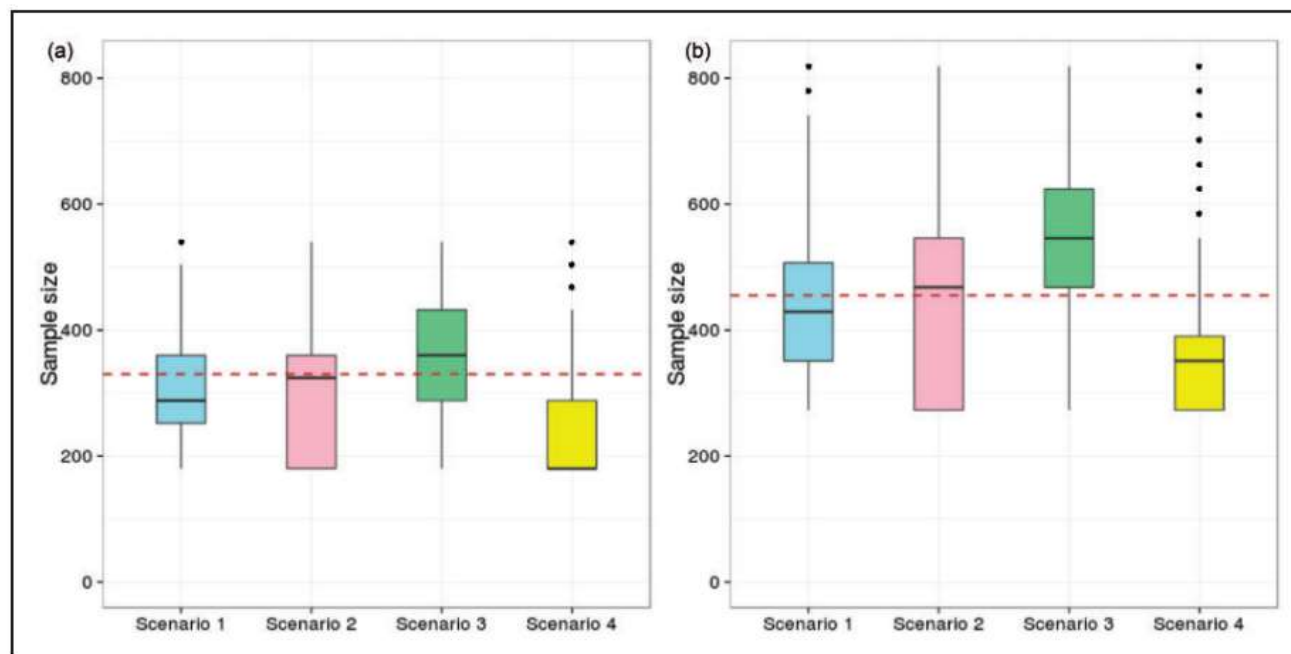


Figure 1. Sample size distribution for three-stage group-sequential MAMS designs with $K=4$ and $K=6$ and four vectors of treatment effects. Scenario 1 – the global null hypothesis (H_G); scenario 2 – the LFC; Scenario 3 – all experimental treatments have uninteresting treatment effect $\delta^{(0)}$; Scenario 4 – all experimental treatments have effect $-\delta^{(0)}$. The dashed red line gives the required sample size for the three-stage drop-the-losers design with the same parameters used: $\alpha = 0.05$, $\beta = 0.1$, $\delta^{(1)} = 0.545$, $\delta^{(0)} = 0.178$.

Survival Endpoints

$$Z_j^{(k)} \approx \frac{\log \widehat{HR}_{kj}}{\sqrt{\frac{4}{d_j^{(o)} + d_j^{(k)}}}} = \log \widehat{HR}_{kj} \cdot \sqrt{\frac{d_j^{(o)} + d_j^{(k)}}{4}}$$

Assuming $k = m$

$p > j \quad (n_p > n_j)$

$$\text{Cov} \left(Z_j^{(k)}, Z_p^{(k)} \right) = \frac{1}{4} \text{Cov} \left(\log \widehat{HR}_{kj} \times \sqrt{d_j^{(o)} + d_j^{(k)}}; \log \widehat{HR}_{kp} \times \sqrt{d_p^{(o)} + d_p^{(k)}} \right)$$

$$\widehat{HR}_{k,p} \approx \widehat{HR}_{kj}^{\left(\frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}} \right)} \times \widehat{HR}_{kj}^{\left(1 - \frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}} \right)}$$

$$\log(\widehat{HR}_{k,p}) \approx \left(\frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}} \right) \log(\widehat{HR}_{kj}) + \left(1 - \frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}} \right) \log(\widehat{HR}_{kj})$$

$Z_j^{(k)}$: Test statistic for arm k at stage j

$\log \widehat{HR}_{kj}$: The logarithm of the estimated hazard ratio (HR) for arm k compared to the control at stage j .

$d_j^{(o)}$ and $d_j^{(k)}$: Number of observed events (deaths or failures) in the control and treatment arm k at stage j , respectively.

$$\begin{aligned} \text{Cov} \left(Z_j^{(k)}, Z_p^{(k)} \right) &= \\ &= \frac{1}{4} \times \text{Cov} \left(\log \widehat{HR}_{kj}, \log \widehat{HR}_{kp} \right) \times \frac{(d_j^{(o)} + d_j^{(k)})^{3/2}}{(d_p^{(o)} + d_p^{(k)})^{1/2}} \\ &= \frac{1}{4} \times \frac{4 \times (d_j^{(o)} + d_j^{(k)})}{d_j^{(o)} + d_j^{(k)}} \times \sqrt{\frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}}} \\ &= \sqrt{\frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}}} \end{aligned}$$

Binary Endpoints

$$Z_j^{(k)} = \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\pi(1-\pi)}}$$

Assume $p > j$ and $n_p > n_j$

We want to solve:

$$\text{Cov} \left(Z_j^{(k)}; Z_p^{(k)} \right)?$$

Notation:

j & p for analyses

k & m for arms

X for Patient's Outcome

$k = m :$

$$\text{Cov} \left(Z_j^{(k)}, Z_p^{(k)} \right) = \sqrt{\frac{n_j}{n_p}}$$

$k \neq m :$

$$\text{Cov} \left(Z_j^{(k)}, Z_p^{(m)} \right) = \frac{1}{2} \cdot \sqrt{\frac{n_j}{n_p}}$$

Objective:

- Effectiveness of "Drop-the-losers" design in Phase II/III oncology trials.
- Type I error rates and statistical power in selecting promising treatments.
- Potential biases under normal distribution assumptions for test statistics.

Scenarios:

- Null Hypothesis: No effect is present.
- Hybrid 1: One effective drug.
- Hybrid 2: Two effective drugs.
- Hybrid 1 + 1: One effective drug and one drug with half the effect size.
- Harmful: One drug is harmful.
- All Effective: All drugs show an effect.

Statistical methods to be evaluated:

- Survival Analysis: Cox Regression, Log-Rank Test.
- [Binary Outcomes: Difference of Proportions, Fisher's Exact Test]

Simulation Study: Survival I

Design Setup (Input)

- Dropout Rate: 10%
- Recruitment Rate: 15 patients / month
- Randomization Block Size: 6
- Control Median PFS: 8 months
- Number of Treatments 4

Drop-the-losers Design: Phase II

- Stage 1: 50 events
- Stage 2: 70 events
- Stage 3: 90 events

Drop-the-losers Design: Phase III

- Stage 1: 100 events
- Stage 2: 200 events
- Stage 3: 400 events

Simulation Setup:

- Number of Simulations: 1'000'000
- 1. Null Hypothesis: No effect
- 2. Hybrid 1: One effective drug
- 3. Hybrid 2: Two effective drugs
- 4. Hybrid 1 + 1: One effective, one half-effective
- 5. Harmful Treatment
- 6. All Effective: All drugs show an effect.

```
# Endpoint simulation (PFS)
PFS <- ifelse(trt == 0, rexp(length(recruit), log(2) / medianPFS_control),
             rexp(n = length(recruit), as.numeric(log(2) / medianPFS_control * Results[i, trt + 2])))
```

Cox-Regression Model:

$$h(t|X) = h_0(t) * e^{(\beta_1 * X_1)}$$

```
coxph(Surv(ad_PFS, event_PFS) ~ ad_trt)
```

Log Rank Test:

$$\chi^2 = \frac{(O_1 - E_1)^2}{V}$$

```
surfdiff(Surv(ad_PFS, event_PFS) ~ ad_trt)
```

Futility Analysis and or drop
the worst test statistics

Parallel Simulation Loop:

- Simulations run in parallel using the package *foreach* loop
- Simulates recruitment, randomization, and endpoint generation
- Censoring

Futility Analysis:

- Calculates HR, test statistic, p-value for each treatment arm.

Only with Cox Regression:

```
Results_sim <- foreach(i = 1:Nsim, .combine = 'rbind', .packages = c('survival'), .inorder = FALSE) %dopar% {  
  Results$Sim <- i  
  simulate_trial(i, thresholds = c(30, 50, 100), Results = Results, RecruitmentRate = 15,  
    patients_stage1 = 150, patients_stage2 = 150, block_size = 6, K = 4, medianPFS_control = 8,  
    dropout_PFS = 0.1)}
```

Combined Cox Regression with Log Rank Test:

```
Results_sim <- foreach(i = 1:Nsim, .combine = 'rbind', .packages = c('survival'), .inorder = FALSE) %dopar% {  
  Results$Sim <- i  
  simulate_trial_log_rank(i, thresholds = c(30, 50, 100), Results = Results, RecruitmentRate = 15,  
    patients_stage1 = 150, patients_stage2 = 150, block_size = 6, K = 4, medianPFS_control = 8,  
    dropout_PFS = 0.1)}
```

Calculation Critical Value c

Set Up the Integral Function:

- Use the `integral_typeIerror()` function to compute the Type I error.

Input Parameters:

- c : Critical value (to be found).
- `mean`: Mean vector under the null hypothesis.
- `var`: Covariance matrix.
- K : Number of treatment arms.
- Desired family-wise error rate (FWER).

Use Numerical Root-Finding:

Find c by solving for when the integral equals the desired FWER using `uniroot()`.

Function:

```
finddesign_survival(J = 3, K = 4, ns = c(4, 2, 1), delta1 = -log(0.7), delta0 = 0, requiredfwer = 0.05,  
                  requiredpower = 0.8, groupsizes = c(1, 2/3, 5/3), events = events)
```

Simulation Study: Survival IV



Decision based on the HR or Test statistic

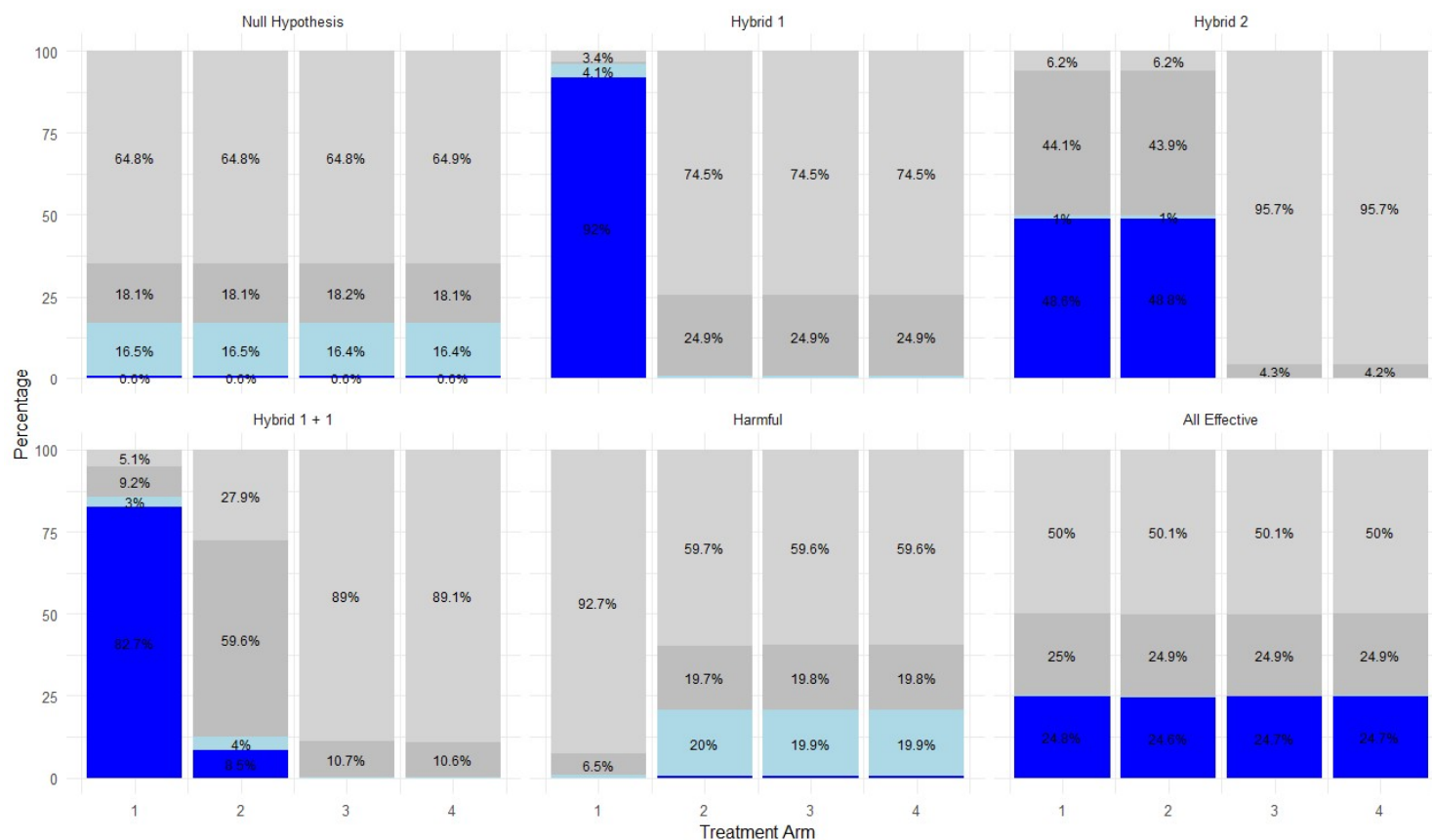
Test statistics from the Cox Regression Model and from the Log Rank Test

Scenario	Treatment Arm	Simulation	AnalysisType	Time Analysis	Outcome	HR	TestStatistics Cox	TestStatistic LogRank	Dropped
1	1	1	Interim 1	27.21366	Continue	0.833 278	-0.90153	0.814976	FALSE
1	2	1	Interim 1	26.75233	Dropped due to Drop-the-Loser Design	0.944 502	-0.56912	0.324246	TRUE
1	3	1	Interim 1	27.08181	Dropped due to Drop-the-Loser Design	0.963 637	-0.55226	0.305297	TRUE
1	4	1	Interim 1	25.36767	Continue	0.966 19	-0.67872	0.461367	FALSE
1	2	1	Interim 2	43.4358	Continue	0.947 042	-0.38425	0.147682	FALSE
1	2	1	Interim 2	41.54692	Dropped due to Drop-the-Loser Design	0.988 072	-0.33736	0.113831	TRUE

Results: Survival I

Outcome Distribution Across Interim Steps and Final Analysis for Phase III

Status Dropped in Interim 1 Dropped in Interim 2 Not Significant in Final Significant in Final



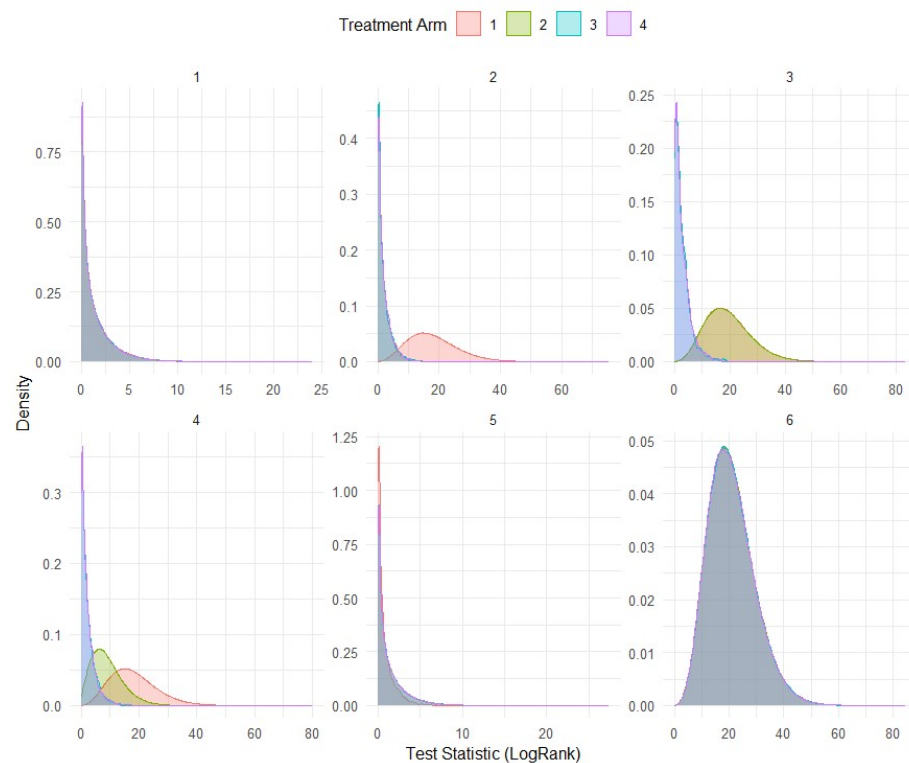
Values smaller than 1% are omitted for readability, except in the null hypothesis plot.

Results: Survival I

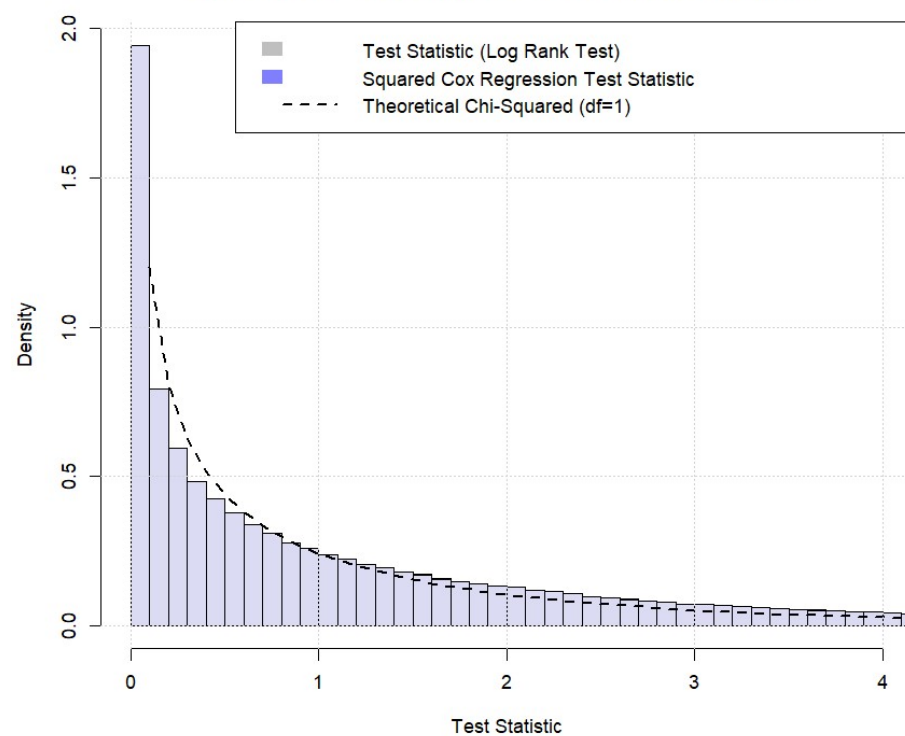
Scenario	Treatment Arm	Dropped in Interim 1	Dropped in Interim 2	Not Significant in Final (95%-CI)	Significant in Final (95%-CI)
1	1	648280 (64.83%)	181359 (18.14%)	16.46% (16.38% - 16.53%)	0.58% (0.56% - 0.59%)
1	2	648027 (64.8%)	180713 (18.07%)	16.54% (16.47% - 16.61%)	0.59% (0.57% - 0.6%)
1	3	647689 (64.77%)	182224 (18.22%)	16.43% (16.36% - 16.5%)	0.58% (0.56% - 0.59%)
1	4	648563 (64.86%)	181473 (18.15%)	16.42% (16.35% - 16.49%)	0.58% (0.56% - 0.59%)
2	1	33595 (3.36%)	5787 (0.58%)	4.1% (4.06% - 4.14%)	91.96% (91.91% - 92.02%)
2	2	744647 (74.46%)	249046 (24.9%)	0.59% (0.58% - 0.61%)	0.04% (0.04% - 0.04%)
2	3	744712 (74.47%)	249087 (24.91%)	0.58% (0.57% - 0.6%)	0.04% (0.03% - 0.04%)
2	4	744730 (74.47%)	248973 (24.9%)	0.59% (0.58% - 0.61%)	0.04% (0.03% - 0.04%)
3	1	61799 (6.18%)	441290 (44.13%)	1.05% (1.03% - 1.07%)	48.64% (48.55% - 48.74%)
3	2	62132 (6.21%)	439388 (43.94%)	1.04% (1.02% - 1.06%)	48.81% (48.71% - 48.91%)
3	3	956709 (95.67%)	42683 (4.27%)	0.05% (0.05% - 0.06%)	0.01% (0.01% - 0.01%)
3	4	956881 (95.69%)	42481 (4.25%)	0.06% (0.05% - 0.06%)	0.01% (0.01% - 0.01%)
4	1	50628 (5.06%)	92278 (9.23%)	3.03% (2.99% - 3.06%)	82.68% (82.61% - 82.76%)
4	2	278586 (27.86%)	596427 (59.64%)	4.01% (3.97% - 4.05%)	8.49% (8.43% - 8.54%)
4	3	889943 (88.99%)	106840 (10.68%)	0.29% (0.28% - 0.3%)	0.03% (0.03% - 0.03%)
4	4	890561 (89.06%)	106281 (10.63%)	0.29% (0.28% - 0.3%)	0.02% (0.02% - 0.03%)
5	1	926554 (92.66%)	64615 (6.46%)	0.88% (0.86% - 0.9%)	0% (0% - 0%)
5	2	596703 (59.67%)	197169 (19.72%)	19.96% (19.88% - 20.04%)	0.65% (0.64% - 0.67%)
5	3	595633 (59.56%)	198425 (19.84%)	19.93% (19.85% - 20.01%)	0.67% (0.65% - 0.68%)
5	4	596159 (59.62%)	197940 (19.79%)	19.94% (19.86% - 20.02%)	0.65% (0.64% - 0.67%)
6	1	499903 (49.99%)	249545 (24.95%)	0.28% (0.27% - 0.29%)	24.78% (24.69% - 24.86%)
6	2	501302 (50.13%)	249445 (24.94%)	0.28% (0.27% - 0.29%)	24.65% (24.56% - 24.73%)
6	3	501227 (50.12%)	248982 (24.9%)	0.28% (0.27% - 0.29%)	24.7% (24.62% - 24.79%)
6	4	500484 (50.05%)	249487 (24.95%)	0.27% (0.26% - 0.28%)	24.73% (24.65% - 24.82%)

Results: Survival II

Distribution of Test Statistics from the Log Rank Test by Scenario and Treatment Arm



Distribution of Test Statistics with Theoretical Chi-Squared



Results: Ethical Aspects

Interim Analysis 2

Scenario	Number of Arms	Total Number of Arms	Probability
1	670368	670368	1.000
2	746307	751793	0.993
3	42481	965778	0.044
4	213026	901268	0.236
5	603767	603767	1.000
6	0	997452	0.000

Final Analysis

Scenario	Number of Arms	Total Number of Arms	Probability
1	681672	681672	1.000
2	18805	979423	0.0192
3	638	996637	0.000640
4	6375	988456	0.00645
5	626802	626802	1.000
6	0	999625	0.000

Extension: Group Sequential Design



$$\begin{array}{c} \text{Futility} \end{array} \left[\begin{array}{cccc} \text{Arm 1} & \text{Arm 2} & \text{Arm 3} & \text{Arm 4} \\ \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \right] \begin{array}{l} > 0 \\ > 0 \\ > 0 \\ > -C_{\text{Futility}} \end{array}$$

The study should be stopped for futility if it falls below the futility threshold.

Arm 1 best at Interim Analysis 1

$$\alpha_1 \begin{array}{c} \text{Futility} \\ \text{GSD} \end{array} \left[\begin{array}{cccc} \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \right] \begin{array}{l} > 0 \\ > 0 \\ > 0 \\ > C_{\text{Futility}} \\ > C_{\text{IA1}} \end{array}$$

We will not stop for futility because the test statistic for the best arm was above the threshold.

Arm 1 best at Interim Analysis 2

$$\alpha_2 \left\{ \begin{array}{l} \left[\begin{array}{cccccccccccc} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > c_{Futility} \\ \left[\begin{array}{cccccccccccc} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > -c_{IA1} \\ \left[\begin{array}{cccccccccccc} 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > c_{IA} \end{array} \right.$$

The test statistic for arm 1 was neither futile nor significant → proceed to IA2.

Final Analysis

$$\alpha_3 \left\{ \begin{array}{l} \left[\begin{array}{cccccccccccc} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > c_{Futility} \\ \left[\begin{array}{cccccccccccc} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > -c_{IA1} \\ \left[\begin{array}{cccccccccccc} 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > 0 \\ \left[\begin{array}{cccccccccccc} 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > -c_{IA2} \\ \left[\begin{array}{cccccccccccc} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] > c_{Final} \end{array} \right.$$

Alpha Spending Function

$$\alpha = \alpha_1 + \alpha_2 + \alpha_3$$

Test whether the test statistic for arm 1 exceeds c_{Final} .

Advancing Clinical Trial Design: Extending "Drop-the-losers" Approach

Key Contribution:

- Extends "Drop-the-losers" to binary and survival endpoints for Phase II/III oncology trials.
- Addresses real-world complexities while maintaining statistical rigor.
- Integrates multiple endpoint types and uses futility analysis (and group sequential design) for earlier decision-making.
- Accelerates treatment selection and drug approval.

Limitations:

- Dynamic endpoint updates (ORR → PFS → OS) add complexity, that correlation at each endpoint's surrogacy needs to be carefully managed.

Ethical Impact:

- Minimizes exposure to ineffective treatments, aligning with beneficence.

Thank you very much for your attention!



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Q&A



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Backup Slides

Results from Wason Paper

Table 1. Sample sizes required for a one-stage design and two-stage and three-stage drop-the-losers designs with $\alpha = 0.05$, $\beta = 0.1$, $\delta^{(1)} = 0.545$ and $\delta^{(0)} = 0.178$.

K	Total sample size required for 90% power			Percentage reduction in sample size	
	$J=1$	$J=2$	$J=3$	$J=1$ to $J=2$	$J=2$ to $J=3$
3	312	282	270	9.6	4.2
4	420	364	330	13.3	9.3
6	637	531	455	16.6	14.3
8	864	715	585	17.2	18.2

Note: For each three-stage design, the number of treatments proceeding to stage 2 is chosen to give the lowest total sample size: in the notation of Section 2, these designs are 3:2:1 for $K=3$, 4:2:1 for $K=4$, 6:3:1 for $K=6$ and 8:3:1 for $K=8$.

Source: Wason et al., 2017

Survival Endpoints

Assuming $k \neq m$

$p > j$ ($n_p > n_j$)

$$\text{Cov} \left(Z_j^{(k)}, Z_p^{(k)} \right) = 0.5 \sqrt{\frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}}}$$